

Databases

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Single table databases

A database is a structured collection of data that allows people to extract information in a way that meets their needs. The data can include text, numbers, pictures; anything that can be stored in a computer. Relational databases will be studied at A Level but for IGCSE only single-table databases will be studied.

A single-table database contains only one table.

Why are databases useful?

Databases prevent problems occurring because:

- if any changes or additions are made it only has to be done once – data is consistent
- the same data is used by everyone
- data is only stored once in relational databases which means no data duplication.

What are databases used for?

To store information about people, for instance:

- patients in a hospital
- pupils at a school.

To store information about things, for instance:

- cars to be sold
- books in a library.

To store information about events, for instance:

- hotel bookings
- results of races.

Fields and records – the building blocks for any database

Inside a database, data is stored in tables, which consists of many records. Each record consists of several fields. The number of records in a table will vary as new records can be added and deleted from a table as required. The number of fields in a table is fixed so each record contains the same number of fields.

An easy way to remember this is: each record is a row in the table and each field is a column in the table.

Note: while databases can contain multiple tables, all the databases considered in this chapter will contain a single table.

Table

Record 1	Field 1	Field 2	Field 3	Field 4
Record 2	Field 1	Field 2	Field 3	Field 4
Record 3	Field 1	Field 2	Field 3	Field 4
Record 4	Field 1	Field 2	Field 3	Field 4
Record 5	Field 1	Field 2	Field 3	Field 4
Record 6	Field 1	Field 2	Field 3	Field 4

Row

Column

A table contains data about one type of item or person or event, and will be given a meaningful name, for example:

- a table of patients called **PATIENT**
- a table of books called **BOOK**
- a table of doctor's appointments called **APPOINTMENT**.

Each record within a table contains data about a single item, person or event, for example:

- Winnie Sing (a hospital patient)
- IGCSE Computer Science (a book)
- 15:45 on January 2020 (an appointment).

As every record contains the same number of fields, each field in a record contains a specific piece of information about the single item, person or event stored in that record. Each field will have a meaningful name to identify the data stored in it, for example:

For a hospital patient the fields could include:

- The patient's first name field called **FirstName**
- The patient's family name field called **FamilyName**
- The patient's date of admission field called **DateOfAdmission**
- The name of the patient's consultant field called **Consultant**
- The patient's ward number field called **WardNumber**
- The patient's bed number field called **BedNumber**, etc.

The PATIENT table structure could look like this:

PATIENT Table

Record 1	FirstName	FamilyName	DateOfAdmission	Consultant	WardNumber	BedNumber
Record 2	FirstName	FamilyName	DateOfAdmission	Consultant	WardNumber	BedNumber
Record 3	FirstName	FamilyName	DateOfAdmission	Consultant	WardNumber	BedNumber
Record 4	FirstName	FamilyName	DateOfAdmission	Consultant	WardNumber	BedNumber
Record 5	FirstName	FamilyName	DateOfAdmission	Consultant	WardNumber	BedNumber
Record 6	FirstName	FamilyName	DateOfAdmission	Consultant	WardNumber	BedNumber

For the table called BOOK the fields could include:

- Title of the book called **Title**
- Author of the book called **Author**
- **ISBN**, etc.

Validation

Some validation checks will be automatically provided by the database management software that is used to construct and maintain the database. Other validation checks need to be set up by the database developer during the construction of the database.

The practical use of a database management system is strongly recommended for all students. The database management software used is *Microsoft Access 365* as Microsoft Access.

For example, the **DateOfAdmission** field will automatically be checked by the software to make sure that any data input is a valid date before it can be stored in the PATIENT table.

Basic data types

There are six basic data types that you need to be able to use in a database:

- text/alphanumeric
- character
- Boolean
- integer
- real
- date/time.

What is a data type?

Each field will require a data type to be selected. A data type classifies how the data is stored, displayed and the operations that can be performed on the stored value. For example, a field with an integer data type is stored and displayed as a whole number and the value stored can be used in calculations.

Syllabus data type	Description	Access data type
text/alphanumeric	A number of characters	short text/long text
character	A single character	short text with a field size of one
Boolean	One of two values: either True or False, 1 or 0, Yes or No	Yes/No
integer	Whole number	number formatted as fixed with zero decimal places
real	A decimal number	number formatted as decimal
date/time	Date and/or time	Date/Time

Primary keys

As each record within a table contains data about a single item, person, or event, it is important to be able to uniquely identify this item. In order to reliably identify an item from the data stored about it in a record there needs to be a field that uniquely identifies the item. This field is called the primary key.

A field that is a primary key must contain data values that are never repeated in the table.

The primary key can be a field that is already used, provided it is unique, for example the ISBN in the book table. The PATIENT table would need an extra field for each record as all of the existing fields could contain repeated data. To create a primary key, we could add a new field to each record, for example a unique number could be added to each patient's record. The extra field is:

Primary key field called HospitalNumber

SQL

Structured Query Language (SQL) is the standard query language for writing scripts to obtain useful information from a database. We will be using SQL to obtain information from single-table databases. This will provide a basic understanding of how to obtain and display only the information required from a database. SQL is pronounced as es-queue-el.

For example, somebody needing to visit a patient would only require the ward number and the bed number of that patient in order to find where they are in the hospital. Whereas a consultant could need a list of the names of all the patients that they care for.

SQL scripts

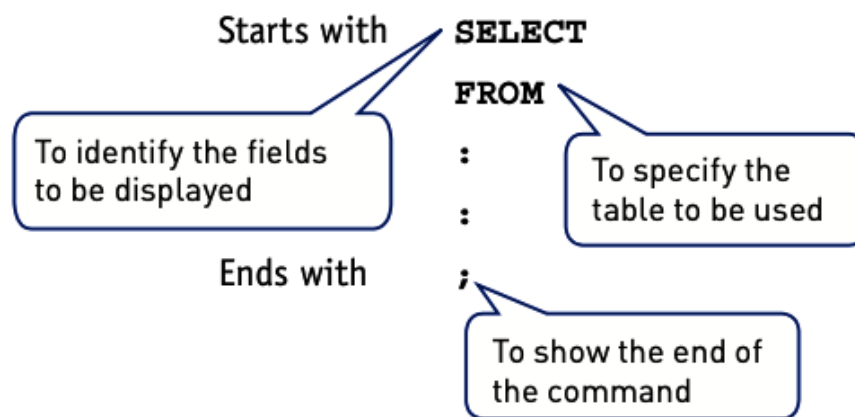
An SQL script is a list of SQL commands that perform a given task, often stored in a file so the script can be reused.

In order to be able to understand SQL and identify the output from an SQL script, you should have practical experience of writing SQL scripts. You can write scripts using SQL commands in Access. There are many other applications that also allow you to do this – MySQL and SQLite are freely available ones. When using any SQL application, it is important that you check the commands available to use as these may differ slightly from those listed in the syllabus and shown below.

You will need to be able to understand and identify the output from the following SQL statements.

SQL Query Statement	Description
SELECT	Fetches specified fields (columns) from a table; queries always begin with SELECT .
FROM	Identifies the table to use.
WHERE	Includes only records (rows) in a query that match a given condition.
ORDER BY	Sorts the results from a query by a given column either alphabetically or numerically.
SUM	Returns the sum of all the values in a field (column). Used with SELECT .
COUNT	Counts the number of records (rows) where the field (column) matches a specified condition. Used with SELECT .

An SQL command:



Only the **SELECT** and **FROM** commands are mandatory in an SQL script. All other commands are optional.

A **SELECT** statement takes the form:

SELECT Field1, Field2, Field3, etc. – this specifies the individual

fields (columns) to be shown.

SELECT * – this specifies that all fields (columns) are to be shown.

A **FROM** statement takes the form:

FROM TableName – this specifies the table to use.

A **WHERE** statement takes the form:

WHERE Condition – this specifies the condition to apply.

Field type	Example value	General notes	Access notes
text	'Mr Smith'	Text field values should be in enclosed in single quotation marks.	Double quotation marks can also be used.
character	'M'	Character field values should be in enclosed in single quotation marks.	Double quotation marks can also be used.
Boolean	TRUE	Boolean can be TRUE or FALSE	Data type is Yes/No
integer	12	Integer field values should be whole numbers.	Allows integer or decimal values.
real	12.01	Real field values should be decimal numbers.	Allows integer or decimal values.
Date/time	'22/11/2022'	Date/time field values should be in enclosed in single quotation marks.	Date/time field values must be in enclosed in hashes (#).

Conditions also require operators to compare values from fields.

Operator	Description
=	equal to
>	greater than
<	less than
>=	greater than or equal to
<=	less than equal to
<>	not equal to
BETWEEN	between a range of two values
LIKE	search for a pattern
IN	specify multiple values
AND	specify multiple conditions that must all be true
OR	specify multiple conditions where one or more conditions must be true
NOT	specify a condition that must be false

An ORDER BY statement takes the form:

ORDER BY Field1, Field2, etc. – this specifies a sort in ascending or

alphabetical order starting with the first field.

ORDER BY Field1, Field2 DESC – this specifies a sort in descending or reverse alphabetical order starting with the first field.

A SUM statement takes the form:

SELECT SUM (Field) – this specifies the field (column) for the calculation.

The field should be integer or real. A COUNT statement takes the form:

SELECT COUNT (Field) – this specifies the field (column) to count if the given criterium is met.

Example

Using the single table database PATIENT you have created.

1. Write an SQL query to list all Mr Jones' patients.
2. Write an SQL query to list all the patients not in ward 6.
3. Write an SQL query to list all the patients who arrived on 12/11/2022.

Practical use of a database

As an OL Computer Science student you need to be able to do the following:

- define a single-table database from given data storage requirements
- choose a suitable primary key for a database table
- read, complete and understand SQL scripts.